11 — Glossary and Project Memory Document

Interpretable Machine Learning for Discovering Hidden Regulatory Switches in Non-Coding DNA

Field	Value
Document ID	GPM-001
Version	1.0
Status	Active — Living Document
Created	2026-06-07
Related docs	Charter, Onboarding

Purpose

This document serves three critical functions:

1. **Glossary**: Defines all recurring terms to ensure consistent language across the team and all documents.
2. **Project Memory**: Preserves decisions already made so future assistants, collaborators, or AI tools do not re-open settled questions.
3. **Scope Guard**: Defines forbidden scope assumptions to prevent drift.

> **This is a living document.** Update it whenever a new term is introduced, a decision is made, or a scope boundary is clarified.

Part 1: Glossary of Project Terms

Biology Terms

Term	Definition as used in this project
Non-coding DNA	DNA that does not encode proteins. Constitutes ~98% of the human genome. Contains regulatory elements that control gene expression.
Regulatory element	A region of non-coding DNA that influences the expression of one or more genes. Includes enhancers, promoters, silencers, and insulators.
Enhancer	A regulatory element that increases transcription of a target gene. Can act over long genomic distances (up to ~1 Mb). Typically marked by H3K27ac and H3K4me1 histone modifications.
Promoter	A regulatory element located near the transcription start site (TSS) of a gene. Initiates transcription. Typically marked by H3K4me3.

Silencer	A regulatory element that represses transcription of target genes.
Insulator	A regulatory element that blocks enhancer-promoter communication or prevents spreading of chromatin states. Often bound by CTCF.
Transcription factor (TF)	A protein that binds to specific DNA sequence motifs and regulates transcription.
Motif	A short, conserved DNA sequence pattern (typically 6–20 bp) recognized by a transcription factor. Represented as a position weight matrix (PWM).
Chromatin	The complex of DNA and histone proteins that packages DNA in the nucleus. "Open" chromatin is accessible and often regulatory; "closed" chromatin is compact and often inactive.
Histone modification	Chemical modifications (methylation, acetylation) to histone proteins that mark regulatory states. Key marks: H3K27ac (active enhancers), H3K4me3 (active promoters), H3K4me1 (poised enhancers).
DNase-seq / ATAC-seq	Assays that measure chromatin accessibility. DNase-seq uses DNase I enzyme; ATAC-seq uses Tn5 transposase. Open regions are "peaks."
ChIP-seq	Chromatin immunoprecipitation followed by sequencing. Identifies where specific proteins (TFs, modified histones) bind to DNA.
TSS	Transcription Start Site — the position where transcription of a gene begins.
cCRE	Candidate Cis-Regulatory Element — a computationally predicted regulatory region defined by ENCODE based on chromatin accessibility and histone marks.
GWAS	Genome-Wide Association Study — identifies genetic variants (SNPs) associated with traits or diseases. Many GWAS hits fall in non-coding regions.
hg38 / GRCh38	The current human reference genome assembly (Genome Reference Consortium Human Build 38).

Project-Specific Terms

Term	Definition as used in this project
Regulatory switch	A sequence region or local sequence-pattern combination in non-coding DNA that appears to strongly influence regulatory activity. This is the project's interpretive framing for biologically meaningful local regulatory patterns. **This is NOT a claim of proven molecular causality.** It is a working label for candidate patterns identified computationally.

Switch-like pattern	A learned or engineered feature (k-mer, motif, CNN filter pattern) that the model identifies as highly predictive of regulatory activity. The "switch" metaphor reflects the idea that specific sequence patterns can toggle regulatory states (active/inactive).
Positive region	A genomic region labeled as regulatory in the project's classification task. Derived from ENCODE cCRE annotations or cell-type-specific peaks.
Negative region	A genomic region labeled as non-regulatory. Sampled from the genome with specific controls (GC-matching, exclusion of known annotations).
Chromosome-holdout split	Data splitting strategy where entire chromosomes are assigned to train, validation, or test sets. Prevents leakage from nearby regulatory elements. Train: chr1–14,X; Val: chr15–18; Test: chr19–22.
Feature importance	A measure of how much a specific feature (k-mer, motif score, etc.) contributes to the model's predictions. Methods: Gini importance, permutation importance, SHAP, saliency.
Motif cross-reference	The process of comparing model-identified important features against databases of known TF binding motifs (JASPAR, HOCOMOCO) to validate biological plausibility.
Kaggle-friendly	A design constraint meaning the system must run within Kaggle notebook limits: 16 GB RAM, 1 GPU (T4/P100, 16 GB VRAM), ~12 hr session, ~20 GB disk, 30 hr/week GPU quota.
Decision gate	A set of criteria that must be met before the project progresses from one modeling phase to the next. Defined in Modeling Roadmap.

Data Terms

Term	Definition
BED file	Tab-delimited text format for genomic intervals: chromosome, start, end, and optional additional columns. Standard format for regulatory annotations.
narrowPeak	Extended BED format used for ChIP-seq and DNase-seq peaks. Includes signal value, p-value, q-value, and peak summit position.
FASTA	Text format for DNA sequences. Each entry has a header line (>name) followed by the sequence.
One-hot encoding	Representation of DNA as a binary matrix: each nucleotide (A, C, G, T) encoded as a 4-dimensional binary vector. A 1 kb sequence becomes a (1000, 4) matrix.
k-mer	A subsequence of length k. k-mer frequency = count of each possible k-length subsequence in a region. For k=4, there are 4^4 = 256 possible k-mers.

PWM / PFM	Position Weight Matrix / Position Frequency Matrix — represents a TF binding motif as a matrix of nucleotide frequencies at each position.
GC content	The fraction of a DNA sequence that is G or C nucleotides. Varies across the genome and between regulatory and non-regulatory regions.

Model Terms

Term	Definition
AUROC	Area Under the Receiver Operating Characteristic curve. Measures classifier performance across all thresholds. 0.5 = random; 1.0 = perfect. Primary metric for this project.
AUPRC	Area Under the Precision-Recall Curve. More informative than AUROC for imbalanced datasets.
SHAP	SHapley Additive exPlanations — a method for explaining individual predictions by computing the contribution of each feature based on cooperative game theory.
Saliency map	A per-input-element attribution map showing which parts of the input most affect the model's output. Computed via gradients for neural networks.
Integrated Gradients	An attribution method that computes feature importance by integrating gradients along a path from a reference input (e.g., all-zeros) to the actual input. More stable than vanilla saliency.
Convolutional filter	A learned pattern in a CNN. First-layer filters on one-hot DNA often learn motif-like patterns.

Part 2: Project Memory — Decisions Already Made

Locked Decisions

These decisions have been made and should NOT be re-opened without documented justification and team consensus.

ID	Decision	Rationale	Date decided	Document reference
D-01	**Project scope is interpretable ML for non-coding regulatory DNA.**	Selected from a broader idea-selection process. Provides clear scientific focus and feasible execution path.	2026-06-07	Charter §10
D-02	**"Regulatory switch" is the project's interpretive label, not a molecular claim.**	Avoids overstatement; keeps language honest and defensible.	2026-06-07	Charter §1, Glossary

D-03	**Initial execution on Kaggle.**	Team lacks institutional compute; Kaggle is free, accessible, and sufficient for Phases 0–2.	2026-06-07	Compute Memo §1
D-04	**Training large genomic transformers from scratch is out of scope.**	Requires multi-GPU, multi-day training costing $10K+; not feasible for a student team.	2026-06-07	Compute Memo §2
D-05	**ENCODE cCREs are the starting dataset.**	Pre-classified, well-curated, freely available, widely used, Kaggle-compatible.	2026-06-07	Dataset Strategy §2
D-06	**Chromosome-holdout is the primary split strategy.**	Prevents leakage from nearby regulatory elements sharing chromatin domains. Train: chr1–14,X; Val: chr15–18; Test: chr19–22.	2026-06-07	Dataset Strategy §6
D-07	**K562 is the primary cell type; GM12878 is secondary.**	Both are well-annotated Tier 1 ENCODE cell lines with extensive data.	2026-06-07	Dataset Strategy §2
D-08	**Interpretability is a first-class requirement, not an afterthought.**	The project's thesis is that interpretable models can discover regulatory patterns. Accuracy without interpretability does not meet the project's goals.	2026-06-07	Charter §4, Research Design §1.4
D-09	**Staged modeling approach (classical → CNN → pretrained).**	Builds complexity incrementally; validates feasibility before investing in expensive experiments.	2026-06-07	Modeling Roadmap
D-10	**Fixed region length of 1 kb for initial experiments.**	Captures most enhancer/promoter-scale regulatory elements; manageable for compute; avoids length-based artifacts.	2026-06-07	Technical Design §2
D-11	**GC-matched negative sampling with exclusion of known annotations.**	Controls for compositional bias; reduces false negatives in negative set.	2026-06-07	Dataset Strategy §4

D-12	**No clinical claims.**	The project produces computational predictions, not medical diagnostics.	2026-06-07	Charter §4 (NG5, NG6)

Open Decisions (Not Yet Resolved)

ID	Question	Options under consideration	Decision needed by
OD-01	Which exact ENCODE experiments (accession IDs) to use for K562?	Multiple DNase-seq and histone ChIP-seq experiments available	Phase 0 data curation
OD-02	Exact k-mer range for baseline features?	k=4 only, k=4+5, k=4+5+6, k=3+4+5+6	Experiment E3 (Phase 1)
OD-03	Whether to include motif scan features in Phase 1 or defer to Phase 2?	Include early (more features) vs. defer (simpler baseline)	Phase 1 planning
OD-04	Which pretrained model to use for Phase 3?	DNABERT-2, Nucleotide Transformer (small), HyenaDNA (small)	Decision Gate 2→3
OD-05	Target publication venue?	MLCB workshop, Bioinformatics journal, bioRxiv preprint	Month 6
OD-06	Whether to build a Gradio demo?	Build demo (outreach value) vs. skip (focus on research)	Month 9
OD-07	License for public code release?	MIT, Apache 2.0, BSD-3	Month 9

Part 3: Forbidden Scope Assumptions

The following assumptions are **explicitly forbidden**. Any contributor (human or AI) must not make these assumptions when working on this project.

#	Forbidden assumption	Why it's forbidden
FA-01	"We have access to a multi-GPU cluster."	We don't. Initial compute is Kaggle free-tier only.
FA-02	"We can train a genomic transformer from scratch."	Requires $10K+ of compute and multi-GPU hardware. Out of scope.
FA-03	"The model's predictions prove molecular causality."	We identify statistical associations, not causal mechanisms.
FA-04	"ENCODE labels are ground truth."	cCREs are computational predictions by the ENCODE consortium, not experimentally validated for every element.

FA-05	"We can validate predictions in a wet lab."	No lab access. Computational validation only.
FA-06	"This is a medical product."	No clinical validation, no regulatory pathway, no diagnostic claims.
FA-07	"We need very long sequences (>10 kb) from the start."	1 kb is sufficient for initial enhancer/promoter-scale analysis. Longer sequences require more compute.
FA-08	"Accuracy is more important than interpretability."	Both matter, but interpretability is the project's differentiating thesis. Don't sacrifice it for marginal accuracy gains.
FA-09	"We should download all of ENCODE."	Start with a curated subset. All of ENCODE is terabytes of data.
FA-10	"Phase 3 and 4 are required deliverables."	They are aspirational extensions. The project is successful if Phase 1–2 produces strong results.

Part 4: Project Labels and Model Classes

Classification Labels

Label name	Code	Definition	Source
Regulatory	`1`	Region annotated as cCRE with active signal in target cell type	ENCODE SCREEN
Non-regulatory	`0`	Randomly sampled region NOT in any cCRE, NOT in coding regions	Generated

Multi-Class Labels (Phase 2)

Label name	Code	Definition
Promoter-Like Signature (PLS)	`pls`	cCRE with high H3K4me3 and DNase near TSS
Proximal Enhancer-Like (pELS)	`pels`	cCRE with high H3K27ac within 2 kb of TSS
Distal Enhancer-Like (dELS)	`dels`	cCRE with high H3K27ac > 2 kb from TSS
CTCF-only	`ctcf`	cCRE with CTCF binding, no other marks
Non-regulatory	`neg`	Generated negative region

Cell Type Identifiers

Cell type	ENCODE tier	Primary use
K562	Tier 1	Primary cell type for all experiments
GM12878	Tier 1	Secondary cell type for cross-cell-type comparison
HepG2	Tier 2	Optional third cell type

Part 5: Acronyms and Abbreviations

Abbreviation	Full form
AUROC	Area Under the Receiver Operating Characteristic
AUPRC	Area Under the Precision-Recall Curve
BED	Browser Extensible Data (genomic interval format)
cCRE	Candidate Cis-Regulatory Element
ChIP-seq	Chromatin Immunoprecipitation followed by Sequencing
CNN	Convolutional Neural Network
CTCF	CCCTC-Binding Factor (a transcription factor)
dELS	Distal Enhancer-Like Signature
DL	Deep Learning
ENCODE	Encyclopedia of DNA Elements
FASTA	Fast-All (sequence file format)
GC	Guanine-Cytosine (nucleotides)
GEO	Gene Expression Omnibus
GTEx	Genotype-Tissue Expression project
GWAS	Genome-Wide Association Study
hg38	Human Genome assembly 38
IG	Integrated Gradients
JASPAR	A database of transcription factor binding profiles

k-mer	A subsequence of length k
LogReg	Logistic Regression
ML	Machine Learning
mm10	Mouse Genome assembly 10
MLOps	Machine Learning Operations
pELS	Proximal Enhancer-Like Signature
PFM	Position Frequency Matrix
PLS	Promoter-Like Signature
PWM	Position Weight Matrix
RF	Random Forest
SHAP	SHapley Additive exPlanations
SNP	Single Nucleotide Polymorphism
TF	Transcription Factor
TSS	Transcription Start Site
XGB	XGBoost (Extreme Gradient Boosting)

Part 6: Update Log

Date	Change	Author
2026-06-07	Document created with initial glossary, decisions, and scope guard.	Project initialization

---

End of Document 11 — Glossary and Project Memory
Next: 12 — Contributor Onboarding Brief